

EXAM PREPARATION

Exercise A:

Find the error for the following commands:

- a) SELECT avg(salary), department_name FROM employees;
- b) SELECT first_name, last_name, department_name, department_id FROM employees CROSS JOIN departments using(department_id);
- c) SELECT first_name, last_name, job_id, job_title, department_id FROM employees NATURAL JOIN jobs WHERE hire_date > 1-Jan-1998
- d) SELECT first_name, last_name, salary+salary*0.1 as „salary“ FOR employees;

Exercise B:

1. Create the **DEPT_EXAM** table with the following structure:

Column	Data Type	Key type
department_id	number(6)	pk
department_name	char(40)	

2. Create the **EMPL_EXAM** table with the following structure:

Column	Data Type	Key type
employee_id	number(6)	pk
first_name	char(30)	
last_name	char(40)	
hire_date	date	
salary	number(8,2)	
manager_id	number(6)	
department_id	number(6)	fk

3. Insert rows into the tables created based on the following tables:

DEPT_EXAM

DEPARTMENT_ID	DEPARTMENT_NAME
50	Shipping
80	Sales
90	Executive

EMPL_EXAM

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	HIRE_DATE	SALARY	MANAGER_ID	DEPARTMENT_ID
100	Steven	King	17-Jun-1987	24000	-	90
101	Neena	Kochhar	21-Sep-1989	17000	100	90
102	Lex	De Haan	13-Jan-1993	17000	100	90
124	Kevin	Mourgos	16-Nov-1999	5800	100	50
141	Trenna	Rajs	17-Oct-1995	3500	124	50
142	Curtis	Davies	29-Jan-1997	3100	124	50
143	Randall	Matos	15-Mar-1998	2600	124	50
144	Peter	Vargas	09-Jul-1998	2500	124	50

Exercise C: For the following tasks use the tables **EMPL_EXAM** and **DEPT_EXAM**:

1. Arrange all the employees using hire_date.
2. Display the minimum salary for employees having department_id=50.
3. Display all the information about the employees having salary between 10000 and 20000.
4. Display the first_name and the last_name in a single column for employees having manager_id=124.
5. Display the first in the alphabetic list of names name for employees having department_id = 50(use the last_name column to arrange)
6. Increase the salary of employees having department_id=50 by 20% and display the salary, the last_name, the first_name and the new salary.
7. Display the average salary for employees where department_id = 90 and round two decimal places.
8. Display the department_name in the DEPT_EXAM table in uppercase.

Exercise D: For the following tasks use the tables **EMPL_EXAM** and **DEPT_EXAM**:

9. Create a join between the tables EMPL_EXAM and DEPT_EXAM and display all the information from both tables for employees having manager_id=124.
10. Create a join between the tables EMPL_EXAM and DEPT_EXAM using department_id and display the first_name, the last_name, the salary, and the department_name for employees having department_id=90.
11. Create a view of all the columns in the EMPL_EXAM table. Add a clause to display the employees having hire_date>01.01.1998.
12. Create a view based on EMPL_EXAM table. Select last_name, first_name, hire_date, department_id for employees who earn more than the average salary from employees having department_id=50.

13. Create a view that contains the first_name, the last_name, and the hire_date for employees having manager_id = 124.
14. Display the first_name, the last_name, and the salary for employees hired after Rajs.
15. Display the first_name, the last_name, and the salary for employees who earn more than the maximum salaries for all the employees having department_id=50.
16. Display the first_name, the last_name, the salary, the department_name and the location_id for employees having the same manager_id as Matos.
17. Display first_name, last_name, and the salary from employees who earn more than the minimum salaries for employees hired after Matos.
18. Create a join between EMPL_EXAM and DEPT_EXAM using department_id and display all the information about employees who don't work in the same department as Rajs. Arrange all the employees using the last_name.

Exercise E: For the following tasks use the UPDATE statement to modify existing rows in the tables **EMPL_EXAM** and **DEPT_EXAM**:

19. Replace the Vargas's salary by 3000.
20. Replace by 3200 the salary for all the employees having the same department_id as Rajs.
21. Update the salary for all the employees having department_id =90 by the average salary for employees having manager_id=124.
22. Update the salary for all the employees hired after Matos by the maximum salaries for employees having department_id=50.

Exercise F: For each of the following questions use tables **EMPLOYEES**, **DEPARTMENTS**, **LOCATIONS**:

23. Create a view based on EMPLOYEES and DEPARTMENTS tables. Select last_name, first_name, hire_date, department_id from EMPLOYEES, the department_name and the location_id from DEPARTMENTS for all the employees having department_id=60. Use department_id to create a join between employees and departments.
24. Create a view based on EMPLOYEES and DEPARTMENTS tables. Select last_name, first_name, hire_date, department_id from EMPLOYEES, the department_name and the location_id from DEPARTMENTS for all the employees who earn more than the average salary from employees having department_id=50. Use department_id to create a join between employees and departments.
25. Create a view based on EMPLOYEES and DEPARTMENTS tables. Select last_name, first_name, hire_date, department_id from EMPLOYEES, the department_name and the location_id from DEPARTMENTS for all the employees hired after Matos. Use department_id to create a join between employees and departments.
26. Create a view to display first_name with capital letters and last_name in one column (from table employees), the department_name, and the location_id from

departments for all the employees having the same manager as Vargas. Use department_id and manager_id to create a join between employees and departments.

27. Create a view to display first_name, last_name from employees, department_name, location_id from departments and the city from locations for all the employees hired after Rajs.
28. Create a view to display first_name, last_name from employees, department_name, location_id from departments and the city from locations for all the employees who earn more than Grant.
29. Create a view of all the columns in the employee's table to display the employees hired before Abel.
30. Create a view that contains the first_name, the last_name, the hire_date, the salary, the department_name and the location_id for employees who earn more than the average salary from department_id=80.
31. Create a view that contains the first_name, the last_name, the hire_date, and the salary for employees having salary higher than the average of the salary from employees having job_id=sa_rep or st_clerk.
32. Create a view that contains the first_name, the last_name, the hire_date, and the salary for employees having the same manager_id as Matos.
33. Create a view that contains the first_name, the last_name, the job_id from employees, the department_name and the location_id from departments for employees having the same job_id as Grant.
34. Create a view based on employee's table. Select last_name, first_name, hire_date, department_id, and employees_id. Display the employees having salary between 4000 and 7000.
35. Create a view that contains the first_name, last_name, and the salary who's last_name begins with "M" or "K".
36. Create a view to display the first_name, the last_name, and the salary for employees who earn less than the maximum salaries for all the employees having manager_id=124.
37. Create a view based on EMPLOYEES and DEPARTMENTS tables. Select last_name, first_name, hire_date, department_id from EMPLOYEES, the department_name and the location_id from DEPARTMENTS for all the employees having the same job_id as Grant. Use department_id to create a join between employees and departments.